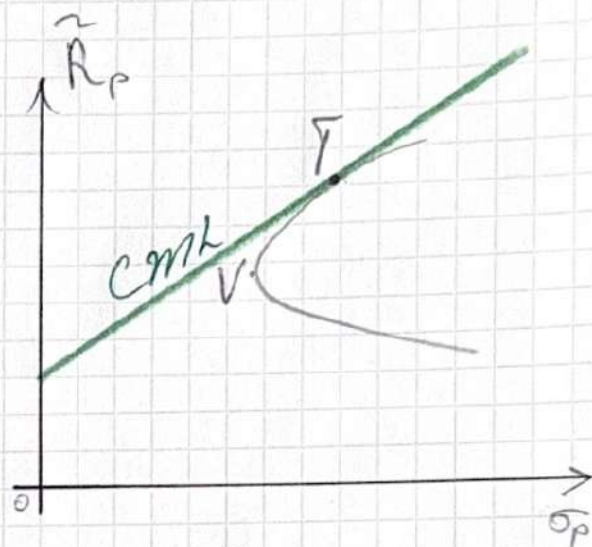


- The risky portfolios alone create the so-called feasible set the interior of which consists of all portfolio combinations including consideration of different weights of portfolios. For a better profitability, investors take into account only the upper half of the set's boundary which is called the efficient frontier and represents all mean-variance portfolios.

- ③. Capital Market Line connects the risk free asset with the tangency portfolio. That's the unique optimal portfolio that contains no investment in the risk-free asset.
- V is the minimum variance portfolio.

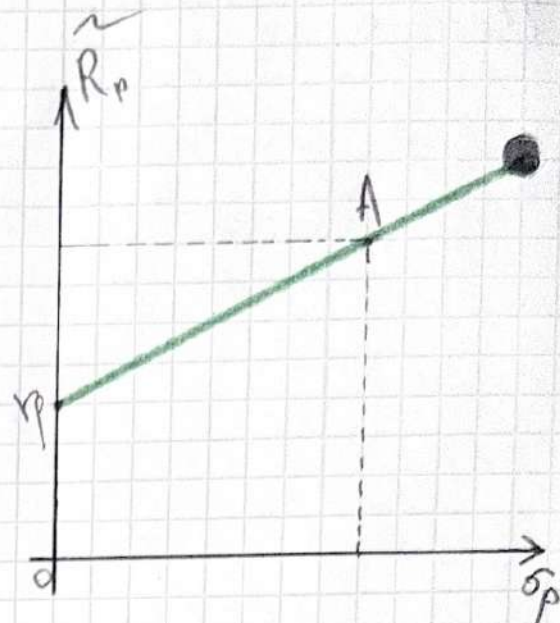


- ④. CML possesses the greatest slope among all other portfolios meaning it has the highest Sharpe-ratio.
- Because of it, the increase of this line is bigger than of any other portfolio. It defines the greatest risk premium in the diagram in the diagram which means investors gain highest reward for any risk taken.
- ⑤ The covariance between each asset and the portfolio must be equal to the assets risk premium.

Mini-case 4.3.

① Assume A is a risky asset and r_f represents the return of a risk-free asset.

- r_f can be found at $\tilde{R}_p$ -axis as its risk is equal to zero, so its σ_p coordinate is zero.



- On the other hand, Asset A is located in the middle of the diagram since its risk value is greater than zero.
- All possible portfolios lay on the straight line as the risk of any portfolio depends on the weight of the asset A .
- The greater is the weight of the asset A in our portfolio, the closer is the point of our portfolio to the asset A on the diagram.

② Since all portfolios in our consideration are risky, we put them somewhere strictly to the right from the $\tilde{R}_p$ -axis.

- r_f is again connected with every risky portfolio by a straight line, as the risk of A, P, W, Q is the only risk the combined portfolios will depend on.

